Practice Test · Class X · Mathematics

Answer Key & Marking Scheme

Step-wise marks · award marks for each correct step even if the final answer is wrong

Time: 1 hr 30 min **Maximum Marks: 40**

SECTION A · 1 mark each

Q	Answer	Q	Answer
1	(a) 4/5	5	(b) 9
2	(c) 1	6	(b) 1
3	(a) $\sec^2 A$	7	(c) $30\sqrt{3}$ m
4	(b) 45°	8	(c) $\sqrt{3}$

SECTION B · 2 marks each

Q9. If $3 \cot A = 4$, find the value of $(1 - \tan^2 A) / (1 + \tan^2 A)$.

$\cot A = 4/3$, so $\tan A = 3/4$	½
$\tan^2 A = 9/16$	½
Substitution: $(1 - 9/16) / (1 + 9/16) = (7/16) / (25/16)$	½
$= 7/25$	½
Total	**2**

Q10. Evaluate: $2 \tan^2 45° + \cos^2 30° - \sin^2 60°$

$\tan 45° = 1$, $\cos 30° = \sqrt{3}/2$, $\sin 60° = \sqrt{3}/2$ (all three values correct)	1
Substitution: $2(1)^2 + (\sqrt{3}/2)^2 - (\sqrt{3}/2)^2 = 2 + 3/4 - 3/4$	½
$= 2$	½
Total	**2**

Q11. If $\sin (A - B) = 1/2$ and $\cos (A + B) = 1/2$, where $0° < A + B \leq 90°$ and $A > B$, find A and B.

$\sin (A - B) = 1/2 = \sin 30° \Rightarrow A - B = 30°$ … (i)	½
$\cos (A + B) = 1/2 = \cos 60° \Rightarrow A + B = 60°$ … (ii)	½
Adding (i) and (ii): $2A = 90° \Rightarrow A = 45°$	½
$B = 60° - 45° = 15°$	½
Total	**2**

Q12. Prove that: $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2 = 2$

$(\sin \theta + \cos \theta)^2 = \sin^2 \theta + \cos^2 \theta + 2 \sin \theta \cos \theta$ and $(\sin \theta - \cos \theta)^2 = \sin^2 \theta + \cos^2 \theta - 2 \sin \theta \cos \theta$	1
Adding: $2(\sin^2 \theta + \cos^2 \theta)$	½
$= 2 \times 1 = 2 =$ RHS (using $\sin^2 \theta + \cos^2 \theta = 1$)	½
Total	**2**

SECTION C · 3 marks each

Q13. Prove that: $(\sec A - \tan A)^2 = (1 - \sin A) / (1 + \sin A)$

LHS = $(1/\cos A - \sin A/\cos A)^2$	**1**
= $(1 - \sin A)^2 / \cos^2 A$	**½**
$\cos^2 A = 1 - \sin^2 A = (1 - \sin A)(1 + \sin A)$	**1**
Cancelling $(1 - \sin A)$: = $(1 - \sin A)/(1 + \sin A)$ = RHS	**½**
Total	**3**

Q14. If $\sec \theta = 13/12$, find the values of all the other trigonometric ratios of θ.

$\cos \theta = 1/\sec \theta = 12/13$	**½**
$\sin \theta = \sqrt{(1 - 144/169)} = \sqrt{(25/169)} = 5/13$	**1**
$\tan \theta = \sin \theta / \cos \theta = 5/12$	**½**
$\cot \theta = 12/5$	**½**
$\operatorname{cosec} \theta = 13/5$	**½**
Total	**3**

Q15. Prove that: $\cos A / (1 + \sin A) + (1 + \sin A) / \cos A = 2 \sec A$

LHS = $[\cos^2 A + (1 + \sin A)^2] / [\cos A (1 + \sin A)]$ (taking LCM)	**1**
Numerator = $\cos^2 A + 1 + 2 \sin A + \sin^2 A$	**½**
= $1 + 1 + 2 \sin A = 2(1 + \sin A)$ (using $\sin^2 A + \cos^2 A = 1$)	**½**
= $2(1 + \sin A) / [\cos A (1 + \sin A)] = 2/\cos A$	**½**
= $2 \sec A$ = RHS	**½**
Total	**3**

Q16. A kite is flying at a height of 60 m above the ground. The string attached to the kite is tied to a point on the ground and makes an angle of 60° with the ground. Assuming there is no slack in the string, find the length of the string.

Correct figure: right triangle, height 60 m, angle 60° at ground, string = hypotenuse L	**½**
$\sin 60° = 60 / L$ (perpendicular/hypotenuse)	**1**
$\sqrt{3}/2 = 60/L \Rightarrow L = 120/\sqrt{3}$	**½**
$L = 40\sqrt{3}$ m (≈ 69.28 m)	**1**
Total	**3**

SECTION D · 4 marks each

Q17. Prove that: $\tan \theta / (1 - \cot \theta) + \cot \theta / (1 - \tan \theta) = 1 + \sec \theta \operatorname{cosec} \theta$

Writing $\cot \theta = 1/\tan \theta$: LHS = $\tan \theta / (1 - 1/\tan \theta) + (1/\tan \theta) / (1 - \tan \theta)$	**1**
= $\tan^2\theta/(\tan \theta - 1) - 1/[\tan \theta (\tan \theta - 1)] = (\tan^3\theta - 1) / [\tan \theta (\tan \theta - 1)]$	**1**
Using $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$: = $(\tan^2\theta + \tan \theta + 1) / \tan \theta$	**1**
= $\tan \theta + 1 + \cot \theta = 1 + (\sin^2\theta + \cos^2\theta)/(\sin \theta \cos \theta) = 1 + \sec \theta \operatorname{cosec} \theta$ = RHS	**1**
Total	**4**

Q18. From the top of a 7 m high building, the angle of elevation of the top of a cable tower is 60° and the angle of depression of its foot is 45°. Determine the height of the tower. (Take $\sqrt{3} = 1.732$)

Correct figure: building AB = 7 m, tower CD, BD = AE = x, angle of depression 45°, elevation 60°	**1**
In $\triangle ABD$: $\tan 45° = AB/BD = 7/x \Rightarrow x = 7$ m	**1**
In $\triangle AEC$: $\tan 60° = CE/AE \Rightarrow CE = 7\sqrt{3}$ m	**1**
Height of tower CD = CE + ED = $7\sqrt{3} + 7 = 7(\sqrt{3} + 1) \approx 19.12$ m	**1**
Total	**4**

Q19. Two poles of equal heights are standing opposite each other on either side of a road, which is 80 m wide. From a point between them on the road, the angles of elevation of the tops of the poles are 60° and 30° respectively. Find the height of the poles and the distances of the point from the poles.

Correct figure: poles AB and CD, height h, point P on road, BP = x, PD = 80 − x	**1**
$\tan 60° = h/x \Rightarrow h = \sqrt{3}\,x$; $\tan 30° = h/(80 - x) \Rightarrow h = (80 - x)/\sqrt{3}$	**1**
$\sqrt{3}\,x = (80 - x)/\sqrt{3} \Rightarrow 3x = 80 - x \Rightarrow x = 20$ m	**1**
$h = 20\sqrt{3}$ m ≈ 34.64 m; distances from poles: 20 m and 60 m	**1**
Total	**4**

Marking notes: Accept any correct alternative method with full marks. In Section D, the figure carries marks only if it is labelled. Do not deduct for not rationalising (e.g. $120/\sqrt{3}$ accepted for $40\sqrt{3}$) unless the question asks for it.